

Mendel University in Brno

Faculty of Business and Economics

SUBJECT :

Text Mining

Supervised by:

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PROJECT :

Sephora Products Reviews Analysis

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PERIOD

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I. Project Objective

The main objective of this project is to analyze some reviews about different sephora products to identify hidden patterns in customer opinions. Using advanced Natural Language Processing (NLP) techniques, and extracting key features such as skin type, ratings…the project aims to identify recurring patterns, consumer sentiment trends, and key product performance indicators. The analysis specifically explores how product perceptions vary across different skin types and rating categories.

II. Data description

The original dataset is composed of multiple CSV files containing over 285,000 unique product reviews. The following core features were extracted and utilized for the analysis:

Feature	Description
review_text	The main body of the user's opinion (Primary text variable).
skin_type	Metadata identifying the user's skin profile (Oily, Dry, Combination, etc.).
rating	Numerical score on a scale of 1 to 5.
is_recommended	Binary indicator of whether the user recommends the product.

III. Loading data

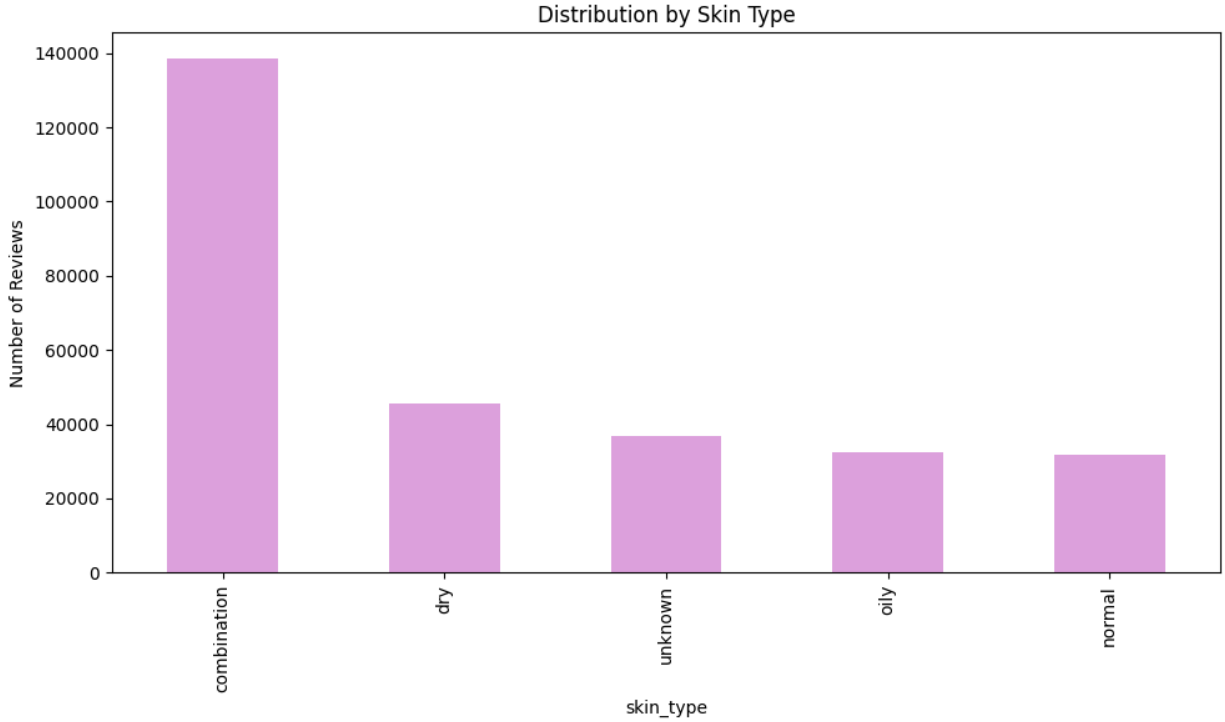
This section handles the initial gathering of all the raw data and getting it ready for analysis. Instead of working with many separate files, I combine them into one clean table.

1. We search for every csv starting with the name reviews and then it merges all those files into one single list (a DataFrame) so I can analyze everything at the same time
2. From all the available columns, we selected only those we are specifically interested in for the text mining analysis, we filtered the dataset to focus on the key columns, such as the review text, skin type, rating, is recommended and product_name.
3. After that, we handled missing values. Since our research depends on text, any row without a 'review_text' was not useful to us, so we decided to remove those records.
4. As a result of this cleaning process, the dataset was reduced from 285,412 to 285,067 records. We successfully filtered out 345 empty entries, ensuring that every row in our final dataset contains the necessary text for a high-quality analysis.
5. To keep as many reviews as possible, we decided not to delete rows with missing values in the 'skin_type' column. Instead, we labeled them as 'unknown' so we could still include

those records in our analysis

6. After labeling the missing values, we plotted the skin type distribution to understand the diversity of users in our study.

```
INFO: Found 5 files. Merging...
INFO: Total records: 285412
INFO: Reviews removed for being empty: 345
INFO: Generating skin type distribution chart...
INFO: 📊 Chart saved as 'skin_type_chart_FINAL.png'
INFO: ✅ Process complete. Data saved to: sephora_cleaned_FINAL.csv
INFO: Final total records: 285067
```



The visualization of skin types provides several key insights into our cleaned dataset of 285,067 reviews:

As shown in the chart, "Combination" skin is the most frequent category by a significant margin, representing nearly 140,000 reviews. This suggests that the majority of Sephora's customer base identifies with this skin type, providing us with a massive amount of text data for this specific segment. Also, the categories for Dry, Oily, and Normal skin types are relatively well-balanced, ranging between 30,000 and 45,000 reviews each. This is an excellent distribution for a Text Mining project, as it ensures we have a large enough sample size to compare how users with different skin needs describe products.

IV. Text Normalization and Preprocessing

The raw textual data in the review_text column is unstructured and contains "noise" that can hinder the performance of machine learning models. To address this, we implemented a sophisticated Natural Language Processing (NLP) pipeline using the SpaCy library. This stage is critical for transforming human language into a standardized format ready for mathematical analysis.

1. *High-Performance NLP Pipeline*

Instead of processing each review individually, we utilized SpaCy's nlp.pipe along with parallel processing (n_process=-1). This allows the script to utilize every available CPU core on the machine, significantly reducing the time required to process over 285,000 reviews. To further optimize efficiency, we disabled the "parser" and "Named Entity Recognition" (NER) components, as they were not required for this specific task.

2. *Normalitation Workflow*

For each review the following steps were applied to extract the essence of the text:

- Tokenization and lowercasting: each sentence is broken down into individual units called tokens. These are automatically converted to lowercase to ensure capitalization doesn't create duplicates.
- Stop Word Filtering: Common words that provide grammar structure but carry little semantic meaning. This focuses the analysis on high-impact keywords.
- Non-alphabetic Removal: We filtered out punctuation,numbers and special characters using the is_alpha attribute. This ensures that only actual words are kept for the final dataset.
- Lemmatization: This process uses a dictionary to find the root of a word (lemma), this reduction is vital for accurately counting word frequencies and identifying core themes.

3. *Re-stringing:*

Once the tokens were preprocessed, they were rejoined into a single string. This result is stored in a new column named cleaned_reviews. This column is the essential part of our dataset, providing a clean and uniform input for subsequent tasks.

V. Feature Extraction (TF-IDF Vectorization)

Once the reviews are cleaned and lemmatized, the next step is converting the text into a numerical format that machine learning algorithms can process. We implemented TF-IDF vectorization using the scikit-learn library.

TF-IDF evaluated how important a world is to a specific review relative to the entire collection

- Term Frequency (TF): Rewards words that appears often ina single review
- Inverse Document Frequency (IDF): Penalizes words that appear too frequently across all reviews.

Parameters:

To ensure the model focuses on the most meaningful terms for the beauty industry, we applied several optimizations:

- Bigrams: We configured the vectorizer to capture two-word phrases, to be more descriptive (skin is less descriptive than dry skink)
- Dimensionality control (max_features): We limited our vocabulary to the top 5.000 most relevant terms. This prevent the dataset from becoming too large
- Frequency filtering: We ignored terms that appeared in fewer than 5 reviews to avoid "noise", and we ignored terms that appeared in more than 90% of the reviews, as they are likely generic words.

Output:

The output is a spare matrix where each row represents a review and each column represents a specific word or bigram. This matrix quantifies the importance of each term and serves as the mathematical foundation for our next steps.

Initially, we performed the TF-IDF vectorization and feature extraction on the entire dataset. However, this global approach revealed a significant limitation known as Lexical Dominance.

When analyzing 285,067 reviews as a single block, high-frequency industry terms like *"skin," "product,"* and *"skin"* dominated the results. While these words are common, they are semantically hollow—they do not provide specific insights into why a product is successful or where it fails for a particular user profile.



To extract "actionable intelligence," we decided to pivot our strategy. We transitioned from a global analysis to a Segmented TF-IDF Approach, crossing two key variables:

- **Skin Type:** To isolate the specific physiological concerns (Dry, Oily, Combination, Normal).
- **Product Context:** To filter out the generic terminology associated with the product category (e.g., removing the word "cleanser" from a cleanser review).

Here we have some examples:

1. Ultra Repair Cream:

Ultra Repair Cream Intense Hydration	combination	2706	moisturizer, dry, face, winter, use
Ultra Repair Cream Intense Hydration	dry	1826	dry, moisturizer, dry skin, face, eczema
Ultra Repair Cream Intense Hydration	oily	375	oily, moisturizer, hand, use, dry
Ultra Repair Cream Intense Hydration	normal	578	moisturizer, hand, dry, face, use

For this product we can see that ‘Dry skin’ users use this product as a treatment for eczemas, while ‘Oily skin’ users perceive the product as too heavy for the face so they use it for their hands.

2. Soy Hydrating Face Cleaner

Soy Hydrating Gentle Face Cleanser	combination	3763	makeup, clean, smell, remove, wash
Soy Hydrating Gentle Face Cleanser	dry	1158	dry, makeup, smell, sensitive, clean
Soy Hydrating Gentle Face Cleanser	oily	736	makeup, clean, smell, remove, wash
Soy Hydrating Gentle Face Cleanser	normal	940	makeup, smell, clean, remove, fresh

In this example we can focus on what are the priorities of the users who want to remove their makeup while ‘dry skin’ users evaluate more that is non-irritating and sensitive with their skin.

VI. Word embeddings

While TF-IDF is excellent for finding unique keywords, it is limited by the "Bag of Words" problem: it treats "moisturizing" and "hydrating" as two completely different things because they are spelled differently. To solve this, we move into Word Embeddings.

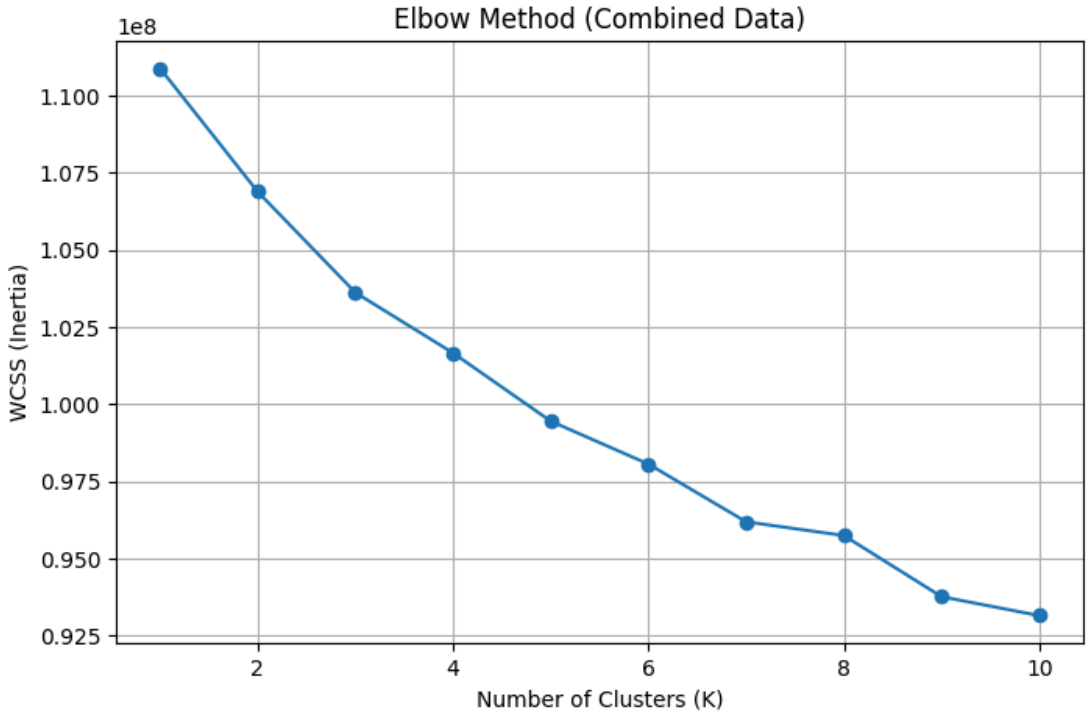
1. Implementation:

To capture the semantic richness of the 285,067 reviews, we utilized a Transformer-based model (all-MiniLM-L6-v2) via the sentence-transformers library. This model transforms raw text into dense numerical vectors of 384 dimensions.

- Each review is encoded into a mathematical coordinate
- The resulting vectors are normalized to ensure that their "direction" represents the meaning, making them ideal for calculating Cosine Similarity.
- These 384 dimensions are "dense," meaning every number carries information about the context.

VII. Determining optimal cluster, The elbow method

To move from semantic representation to Automatic Categorization, we performed a Clustering analysis. The first challenge in unsupervised learning is determining the optimal number of groups (K) without prior labels.



Following the determination of the optimal K=6, we implemented a K-Means algorithm on a fused data structure. To uncover the common latent patterns hypothesized in the project scope, we concatenated:

1. **Semantic Vectors:** The 384-dimensional text embeddings.
2. **Categorical Metadata:** One-Hot Encoded variables for *Skin Type* and *Product Name*.

By scaling these features and projecting them into a unified space, the model identifies clusters that are not only linguistically similar but also demographically and contextually related. This approach is fundamental for identifying why certain negative experiences are mathematically grouped around specific skin profiles.

1. **Technical Implementation: The Fusion Process**

The script follows a three-step pipeline to create a unified mathematical space:

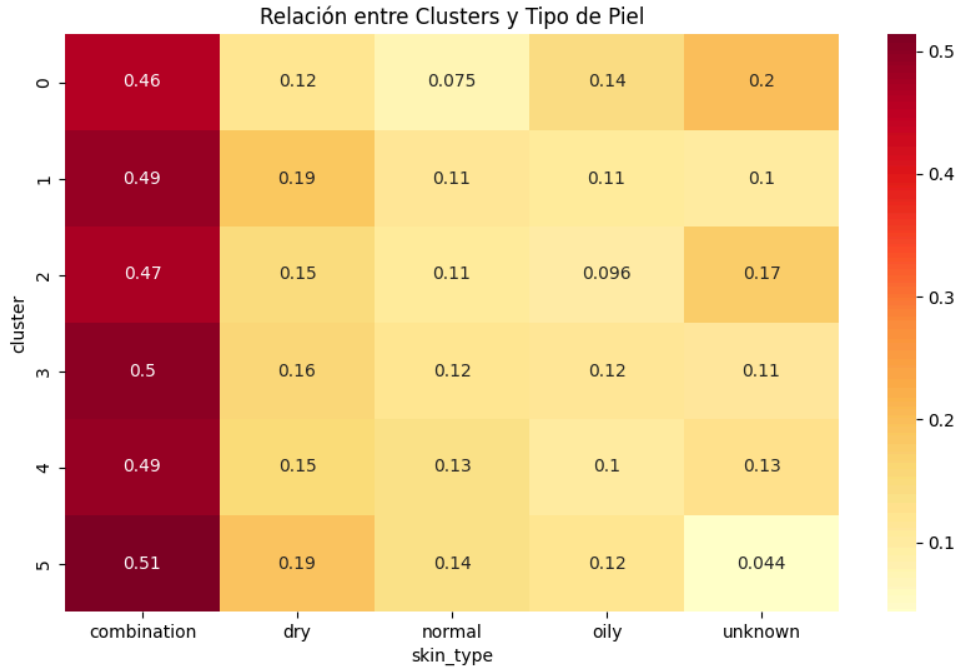
- **One-Hot Encoding (OHE):** Categorical variables, specifically skin_type and product_name, were transformed into binary vectors. This allows the model to process non-numeric data alongside the text.
- **Standard Scaling:** Since embeddings and categorical codes have different scales, we applied a StandardScaler. This ensures that a specific skin type (like "Sensitive") carries the same mathematical weight as a sentiment-rich word (like "Irritation").

2. **The output:**

Skin heatmap

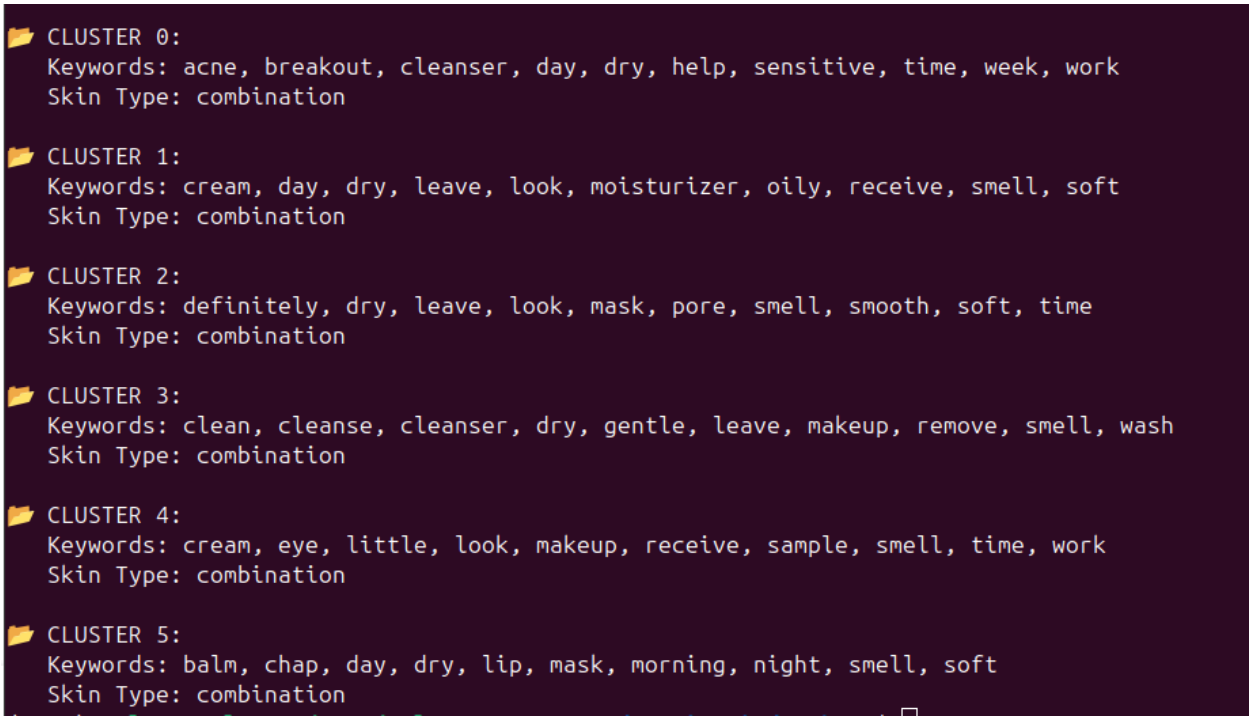
Once the clusters were organized, we made a multi-dimensional analysis for each group. This process was structured around three core pillars:

First we examined the demographic composition by analyzing the distribution of skin types in each cluster, second, the most frequently used products were identified to understand the commercial context of each group. Finally, we analyze the review to understand the sentiment and concerns of the users.



The skin type heatmap illustrates the distribution of declared skin types across the six clusters, providing a dermatological signature for each group. The most prominent feature is the high intensity in the “Combination” column across all clusters; this reflects the real-world market demographic of Sephora customers. But the model strength lies in the ability to detect variance in the minority groups, which are the true drivers of semantic separation.

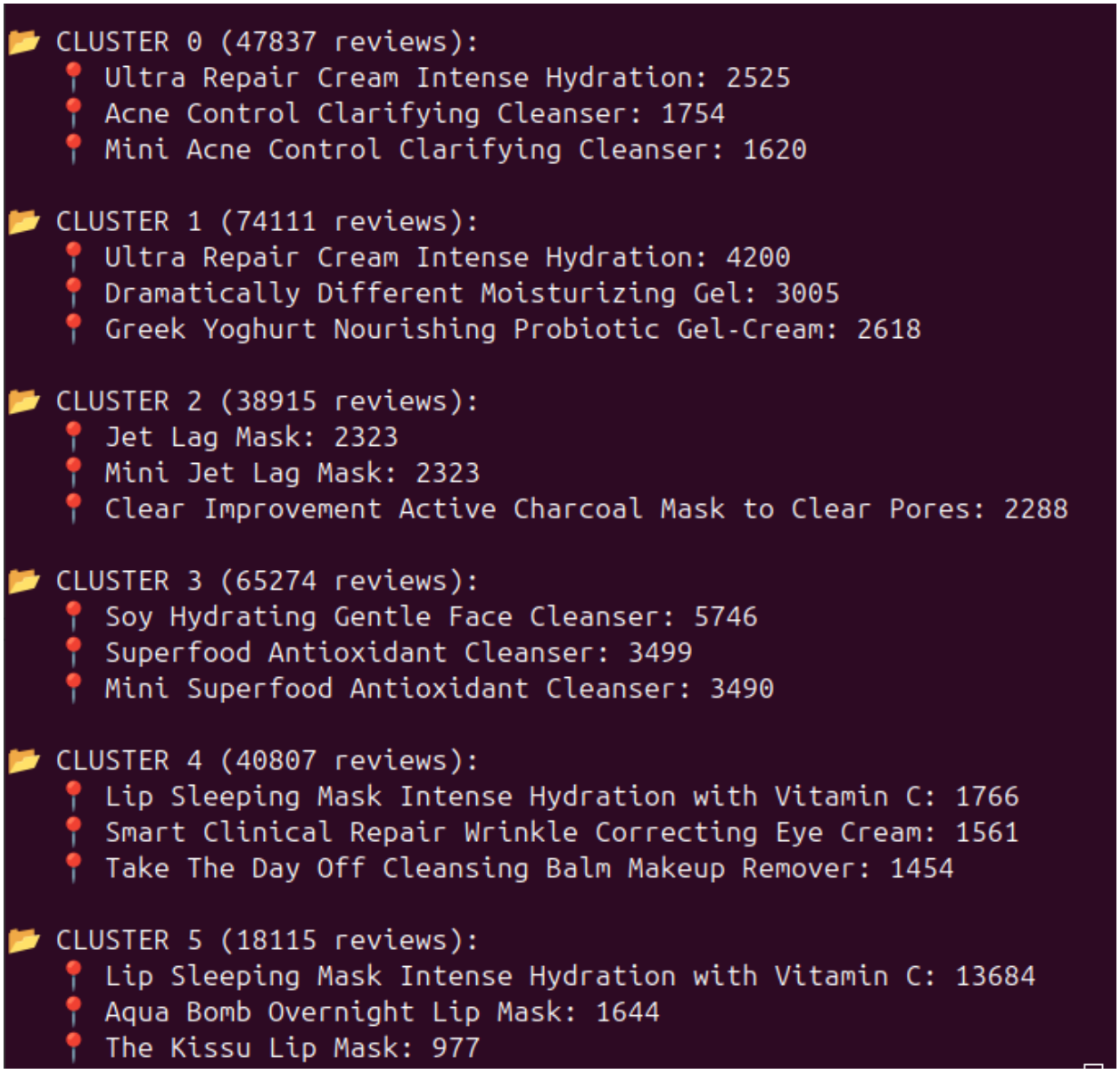
Words frequency of each cluster:



If we look at the review analysis we can find different patters:

- Cluster 0: This cluster exhibits a higher percentage of negative sentiment compared to the rest of the dataset. This cluster contains words like acne or breakout which has a negative sentiment, this cluster is not discussing the satisfaction of the product, but rather about physical reactions. It's a crisis zone for brands, they could find in this cluster things to look after to improve in their products.
- Cluster 1, 2 & 3: This cluster has a strong correlation between sensory attributes and customer satisfaction, we can see this in words like, smell,soft,moisturizer… Users judge on their efficacy (cleaning, removing)
- Cluster 4,5: These are high satisfaction clusters focusing on customers that tried eye or lips products, we see words like works, day, night which give us information about the performance of the products.

Top products in each cluster:



- Cluster 0: Acne & Hydratation products
- Cluster 1: General Hydratation
- Cluster 2: Facial Mask
- Cluster 3: Cleaning products
- Cluster 4: Anti-Aging / Eyes
- Cluster 5: Lip Specialist

VIII. Summarize

The final conclusion of this project focuses on the transition from massive data processing to evidence-based decision making. Based on the patterns identified through the fusion of semantic vectors and metadata, three fundamental pillars for the cosmetic industry emerge:

The analysis of Cluster 0 serves as an early warning system for brands. By identifying terms such as "acne" or "breakout" associated with specific hydration products, companies can take proactive measures:

- Product Reformulation: Investigating whether specific active ingredients are causing adverse physical reactions in certain skin profiles.

This research proves that the "average consumer" profile is insufficient for high-level analysis. The Segmented TF-IDF approach allows for real-world personalization:

- Niche Marketing: The analysis revealed that while "Dry skin" users treat the *Ultra Repair Cream* for eczema, "Oily skin" users find it too heavy for the face and use it for their hands instead. Brands can capitalize on this by promoting the product as an intensive hand/body treatment for oily-skin segments.
- Tailored Value Propositions: For products like the *Soy Face Cleanser*, brands should emphasize "non-irritating" and "sensitive" qualities for dry-skin users, while highlighting "makeup removal" and "deep cleaning" for other segments.

The success of Clusters 4 and 5 (eyes and lips) provides a roadmap for "best practices". By observing terms like "works," "day," and "night," it is confirmed that consumers value tangible performance and temporal functionality:

- Since keywords like "sample" and "receive" appear in high-satisfaction clusters, it validates that product trial strategies are highly effective for generating positive sentiment in specialized categories.

In conclusion, this project transcends traditional word frequency analysis by leveraging advanced Natural Language Processing (NLP) and K-Means clustering to extract actionable market intelligence. By identifying how product perceptions vary across specific customer profiles such as the distinct usage patterns of 'Oily' versus 'Dry' skin type, this study provides a dermatological signature of the Sephora consumer base. These insights enable brands to refine product formulations, address 'crisis zones' like physical skin reactions, and execute hyper-segmented marketing strategies that align with the real-world needs of diverse user profiles.